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Sentiment Analysis of Hotel Reviews - a Comparative Study

Gauthami Sreenivas
Department of Computer Science
PES University
Bangalore, India
gauthamisreenivas@gmail.com

Kishan Minna Murthy
Department of Computer Science
PES University
Bangalore, India
kishanminna@gmail.com

Kshitij Prit Gopali
Department of Computer Science
PES University
Bangalore, India
gopalikshitij@gmail.com

Navya Eedula
Department of Computer Science
PES University
Bangalore, India
navyaeedula@gmail.com

Dr.Mamatha H R
Department of Computer Science
PES University
Bangalore, India
mamathahr@pes.edu

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***Abstract*—Sentiment analysis is an important domain in Natural Language Processing (NLP) since it is an efficient way to extract features and user sentiments from textual data. Performing sentiment analysis of big data in the tourism industry is useful for businesses to understand the needs of their customers and improve hotel facilities to increase customer satisfaction. This paper aims to compare, analyze and employ different types of supervised, unsupervised, and pre-trained models. The supervised models - Decision Trees, XGBoost, Multinomial Naïve Bayes, Multinomial Logistic Regression, SVM, and Stochastic Gradient Descent were tested and the parameters were optimised using GridSearchCV. Two unsupervised models, K-means clustering and Latent Dirichlet Allocation were implemented with TF-IDF and Word2Vec embeddings. The pre-trained models, VADER and TextBlob were also implemented. The labelled dataset used for this study contains user reviews of hotels around the world, where each review is classified as positive, neutral, or negative. The SVM model resulted in the highest weighted F1 score of 0.8516.**

***Index Terms*—Sentiment Analysis, Hotel Review Data, SVM, Decision Trees, Multinomial Naïve Bayes, Stochastic Gradient Descent, XGBoost, K-means, LDA, VADER, TextBlob, TF-IDF, Word2Vec**

I. Introduction

Text classification algorithms have widely been used for many applications including optimizing search engines, processing e-mails, extracting features from social media data, and analyzing customer buying trends. These tasks are possible with the recent advancement in Natural Language Processing (NLP) through which analytical model-building can be computerized and insights can be drawn about real-world data based on processing human languages. Text classification techniques are used to structure, organize and categorize text data that can be obtained from sources like websites, documents, social media, audio transcripts, etc.

Working with text classification proves to be challenging especially due to the nature of the data. Unstructured data in the form of reports, e-mails, news, etc, requires in-depth analysis to discover hidden relationships and meaningful patterns between the entities. Unstructured data often varies in terms of quality and format. Hence, more sophisticated techniques like sentiment analysis, pattern recognition, prescriptive analytics, and cognitive analytics, are required for better analysing unstructured data. The recent advancements in technology led to inexpensive computational processing and data storage systems which have led to faster analysis of complex data via automatic data models.

Sentiment analysis which is a domain of text analysis, employs machine learning techniques to analyse sentiments, emotions, opinions, attitudes, judgements, and evaluations on entities like services, organizations, problems, events, products, individuals, topics and their attributes. Applications of sentiment analysis have been widely used in almost all fields from businesses and health care to online retail and financial services. It proves to be an efficient way to understand and infer from customer feedback and public opinion. Businesses analyse this data and determine positive or negative sentiments associated with the brand. Companies can aim to cater to the needs of their stakeholders better by improving their services according to the received feedback and in turn promote customer loyalty.

II. Literature Survey

An approach proposed in [1] suggests a method to analyse travel reviews on TripAdvisor by extracting keywords from these reviews and obtaining a concept list of keywords using the Microsoft Knowledge Graph. The study experiments with TF-IDF, LDA and Word2Vec word embeddings models. It was found that Word2Vec model achieved excellent performance in both text classification and similarity calculation. Three variations of the SVM model were used for the classification task - SVM, SVM with Word2Vec embeddings and Random Over-Sampling in combination with the Knowledge graph. Their F1 scores were 0.148, 0.830 and 0.879 respectively.

Authors in [2] propose sentiment analysis on the hotel reviews dataset and additionally an aspect-based categorization of user reviews. The three sentiments, positive, neutral and negative are considered from a diverse dataset of hotel reviews that have been scraped from Tripadvisor. The model built is an ensemble of the infamous, BERT model in combination with a Random Forest-based classification model in order to capture different textual features. The BERT pre-trained model with Random Forest classifier has been used with a weight-assigning protocol. The model produced an accuracy of 94.35%, 92.79%, and 92.36% on the training, validation and testing datasets, respectively.

Authors in [3] aim to extract certain textual features from hotel reviews about specific products or amenities. The data is scraped from TripAdvisor, online panel surveys and from text messages. The research aims to compare the performance and results of supervised and pre-trained models in comparison to human classifiers. SentiStrength and DeeplyMoving, which were the two pre-trained classifiers used, had varying accuracies depending on the dataset while the supervised learning models, SVM and Naïve Bayes, were consistently accurate in all datasets with accuracies varying between 0.85 to 0.89. The authors conclude that hyper-parameter tuning is an efficient way to make the supervised models more efficient while pre-trained models need to be used as "black-box models" with no way to change the parameters even if they perform poorly on large and noisy datasets.

[4] proposes to solve the problem of high-dimensional low-volume hotel review text data. It demonstrates how to represent text feature vectors using Word2Vec and ISODATA clustering algorithms. The Word2Vec model is trained on a large dataset of hotel reviews and then passed to the ISODATA clustering algorithm. This algorithm groups the words based on the distance between the word vectors. The accuracy reported was 0.25% higher than that of text feature vector representation based on Word2Vec and the K-means clustering method for sentimental analysis of hotel reviews . 0.31% increase was observed on the AUC value. In [5], the authors aimed to conduct a study on the efficiency of using sentiment analysis on text reviews of movies, products from Amazon and editorials from NYTimes. The study used the pre-trained VADER model. The authors then applied a threshold based on polarity values. It proved to be the most accurate when compared with human classification. The paper also comments on the trend of ratings left by the users. It was found that a majority of them were 5-star and 4-star reviews in comparison to negative reviews. A neutral rating of 3 was the least in number. Another analysis of reviews is presented in [6]. This study analyzes and categorizes customer reviews for Android apps using supervised learning models. It was shown that the SVM model (with non-linear kernel) and Random Forest Classifier gave accuracies in the range of 43-45%. The Naïve Bayes model was proven to be the best with an accuracy of 58.5%. The size of the data was increased from 5,000 to 75,000 reviews per class and it was proven that there was no significant growth in the accuracy of the model (about 1-2%). [7] presents a study on online reviews. The sentiment classification methods were selected based on the text documents. The semantic analysis of sentiment words was also proposed. The Word2Vec model was used to retrieve word vectors and the LDA model was used to retrieve the topic vectors of the review text documents. Using two types of semantic similarity, it is considered beneficial to select the word with the greatest similarity as the primary classifier for text documents.

[8] performs a text classification on the 20newsgroups dataset consisting of 20000 newsgroup emails. A total of 190 classification tasks were examined by studying the set of one-against-one binary classifications by utilizing the models SVM, KNN and Naïve Bayes. It is stated that the Naïve Bayes algorithm achieves good performance for classification and is significantly faster than SVM. The paper points out that the clear drawback of SVM is the large amount of time and resources it takes for training a large dataset as it increases linearly with the number of features added. [9], [10] used SGD optimization on the classical SVM model combined with TF-IDF vectorization. The paper [11] proposes SGD Optimization over classification models like logistic regression, perceptron classifier and SVM with subsequent hyper-parameter tuning using Grid-Search to enhance the accuracy. The authors advocate the use of hyper-parameters before training the classifiers in order to improve the classification technique. The model performance is measured using the traditional metrics of accuracy, precision, recall and F1-score which were 0.848, 0.844, 0.848 and 0.819, respectively for the SVM model.

In paper [12], K-Means and Naïve Bayes algorithms are chosen and used to analyze sentiments or public from social websites. Additionally, the removal of outliers was also added. It was seen that the combination of K-Means and Naïve Bayes algorithms had lesser accuracy than Naïve Bayes algorithm without the combination of K-Means clustering. In the first case, the Naïve Bayes model was trained on labels generated by the K-means algorithm while in the second case it was trained based on the labels generated by Sentiwordnet. The accuracy of the combination of Naïve Bayes and K-Means ranged from 80.3%-81.5% while the accuracy of the Naïve Bayes algorithm on its own ranged from 80.5%-82.5%. [13] presents a study to analyze social media data and perform a supervised classification of Turkish Texts into various categories - like Sports, Health, Technology, etc by using Naïve Bayes, KNN, Decision Trees and SVM models. The Multinomial Naïve Bayes model gives an overall accuracy of 90%. Subsequently, k-fold cross-validation, with value $k = 5$, was implemented to assess the accuracy rates of all the classifiers.

The study put forth by [14] conducts a sentiment analysis on film reviews but in a fine-grained manner to evaluate the viewer's sentimental orientation and intensity in different aspects of the film, i.e., acting, direction, etc. using the Amazon dataset. XGBoost proved to be the model with the highest accuracy of 0.885 and best F1 score of 0.8713. In the study presented by [15], specific data and information such as user reviews of popular movies have been analyzed using data visualizations and the LDA theme analysis model to develop trends of movies based on user preferences.

Various studies were proposed to perform textual analysis of tourism data. In [16], an aspect-based sentiment analysis of user reviews on Google Maps of tourism destinations in Indonesia was conducted using supervised machine learning algorithms. A textual sentiment classification model was built, and various sentiment features were combined to construct a multi-input matrix. This was passed as an input to a multichannel CNN model in order to extract sentiment features and complete the textual sentiment classification([17]). Another model, DomBERT, an extension of BERT is applied to end tasks in aspect-based sentiment analysis to obtain results that were promising. Another paper aims to learn domain-oriented language models and combine general language models, i.e.., BERT and ELMo, as well as domain-specific language understanding [18].

III. Motivation

Although sentiment analysis is a widely explored domain in text classification, its use in hotel and travel data is quite limited. User ratings are scalar in nature and do not provide any information about the specific amenities they like/dislike in the hotel. Such information can be found only from feature extraction of textual reviews. These opinions can prove useful to future visitors of the hotel or for the hotel management to improve their services. There are various techniques and models to perform sentiment analysis. This involves usage of various pre-processing techniques, text feature extraction models and machine learning algorithms to classify the reviews into the correct category. Often deciding the correct model or pipeline is tedious task since different types of data need different techniques. Thus, we aim to provide a thorough analysis of supervised, unsupervised and pre-trained models to determine which works best on the labelled dataset considered in the study. We also aim to explain the features of the data that resulted in the high or low performance of each model.

IV. Dataset Features

The dataset utilized in this study is taken from IEEE Dataport [19]. It consists of two files, namely, dataset_finale.tsv and Sentiments.tsv, where the latter consists of 58612 English textual reviews from hotels all over the world and their corresponding sentiment which can be negative, neutral or positive. It is a skewed dataset consisting of 46738 positive, 7423 neutral and 4451 negative reviews.

V. Methodology

Fig.1 provides high-level understanding of the workflow. It depicts the three stages - Data Preparing and Pre-processing, Multi-class Classification Models, and Model Evaluation.

A. Data Preparing and Pre-processing

The review data was pre-processed using many functions present in the standard NLTK library and is the first stage depicted in the Fig.1. This was done to improve the accuracy of the training models, reduce the noise in the dataset, and lessen the burden on computational resources. The following pre-processing techniques were employed sequentially -

- **Removal of non-English reviews** by using functions from the 'langdetect' Python library
- **Data Cleaning** which includes the removal of URLs, digits, special characters and punctuation marks
- **Tokenization:** Separating sentences into a list of words
- **Case-folding:** Transforming all letters to lowercase
- Removal of **stopwords** like 'a', 'then', 'this' which do not hold significant sentiment values
- **Lemmatization:** Reducing every word to its root word
- **Stemming:** Transforming every word into its word stem by removing prefixes and suffixes

After performing the necessary pre-processing, the dataset consisted of 4446 negative, 7410 neutral and 46685 positive reviews. A 70-30 train-test split was used for training and evaluation of all the models. The amount of time taken for training is also taken into consideration in this study.

B. Multi-class Classification Models

For the supervised learning models, an ML pipeline is built which first converts the pre-processed text data to word vectors using the word embeddings models (by sequentially employing CountVectorizer and TfidfVectorizer) and then classifies the reviews into one of three classes. Unsupervised models were experimented with both the TfidfVectorizer and Word2Vec models to analyze which of the two gave better performance. The pre-trained models were tested with two types of inputs - the raw text from the dataset and the pre-processed data input to understand if the pre-processing techniques affect the performance of these models.

1) Models for Word Embeddings: Word Embeddings are numeric vector representations of text in a lower-dimensional space which are then passed as input to ML models. These models perform feature extraction preserving the syntactical and semantic information of the textual data.

a) Count Vectorizer: Count Vectorizer converts the text corpus into a sparse matrix which contains the frequency of occurrence of each word.

b) TF-IDF Word Embeddings: TF-IDF ranking solves the problem of higher frequency terms having more importance and hence gives an accurate representation of the frequency of keywords present in the corpus([9], [10]). TfidfVectorizer combines CountVectorizer and TfidfTransformer in a single model.

c) Word2Vec Word Embeddings: The Word2Vec model first converts words to vectors and then calculates the distance between these vectors. The larger the distance, the higher the similarity between the two words. This method effectively expresses the words in high-dimensional vector form [1].

2) Supervised Learning Models: The models are trained with the dataset having labels - positive, negative and neutral. The attributes of each labelled class are automatically learnt by the model. The following models were tested with the

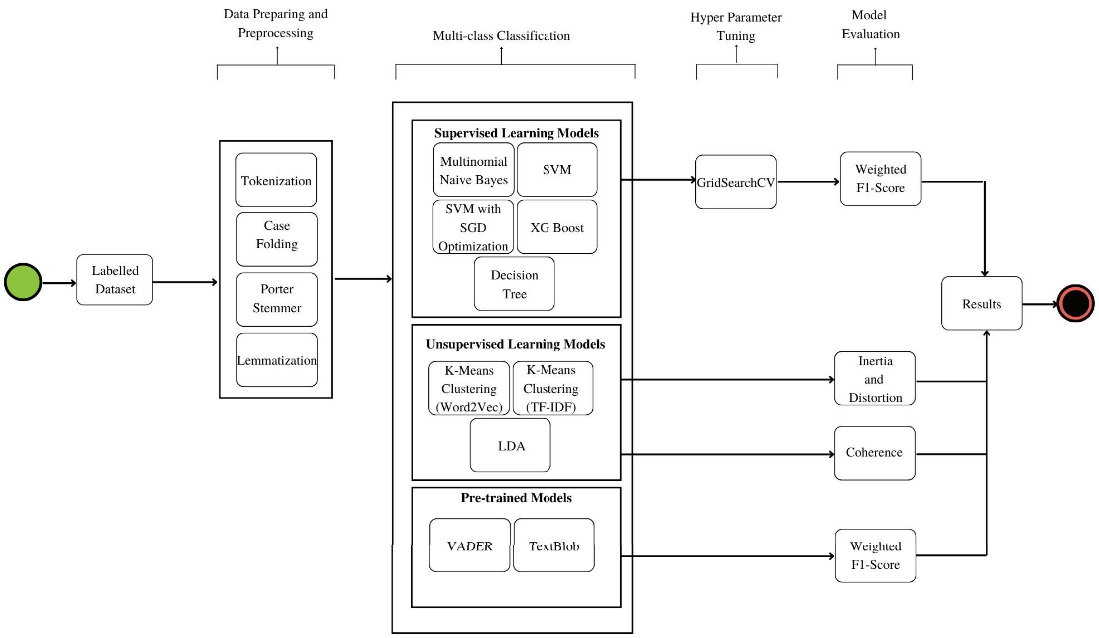

Fig. 1. Workflow Model

dataset in the study. To determine the optimal parameters, hyper-parameter tuning was performed using GridSearchCV which exhaustively generates all the candidates from the grid to determine the best parameters.

a) Decision Trees: The decision tree is a tree-structured classifier. The internal nodes represent the features of the dataset, branches represent the decision-making process, while each leaf node is the classification result. The algorithm begins at the root node. Ech of the values of the record attribute and the root attribute are compared before moving on to subsequent nodes. Although decision trees are useful when the data contains a few highly relevant attributes, the computational complexity significantly increases with the increase in complexity amongst the attributes' relationships. The Gini Index is used as the cost function and is given by -

$$Gini = 1 - \Sigma_{i=1}^{C} (p_i)^2 \tag{1}$$

where,
p_i = probability of an event i of a class
C = total number of classes

b) Multinomial Naïve Bayes: This algorithm based on the Bayes theorem, is a probabilistic classifier which makes predictions based on the likelihood that an event will occur. It makes an assumption that the occurrence of one trait is independent of the occurrence of other features. This is why the algorithm is known as *"Naïve"*. Using conditional probability, the likelihood of a hypothesis is calculated using the Bayes theorem. The Naïve Bayes model is strongly influenced by the conditional independence assumption between the different attributes. As the number of features increases, the performance of the model is said to increase without a significant increase in training time.

The conditional probability for Bayes theorem is given by -

$$P(C|X) = \frac{P(X|C).P(C)}{P(X)} \tag{2}$$

where,
C = class of possible outcomes
X = instance that needs to be classified

c) Multinomial Logistic Regression: The logistic regression classifier is a model that generalises the regression to a classification problem and gives two or more classes as the output. The maximum likelihood estimator is used to evaluate the probability of the class of a dependent variable based on multiple independent variables. A linear relationship between the dependent and independent variables is determined and a sigmoid function is applied over this relationship to introduce non-linearity [20]. The equation for the MLR model is as follows -

$$log(p_j(x)) = e^{\beta_{0j} + \beta_{ij} X_1 + \ldots + \beta_{pj} X_p} p_J(x) \tag{3}$$

where,
p_j = probability of class belonging to j
p_J = reference probability of class belonging to J
J = Total number of categorical variables
j = 1,...,J-1

d) Support Vector Machines: This model performs the classification task by first identifying a decision boundary, known as a hyperplane, in an N-dimensional space which categorizes the data points into different classes. The number of features determines the hyperplane's size. This hyperplane that most accurately depicts the distance between the classes is considered to be the best one and the hyperplane whose distance to the closest data point on either side is maximum is selected. SVM is exceptionally resistant to outliers. The SVM kernel technique is a function that converts a distinct problem from a low-dimensional input space into a higher-dimensional space.

The SVM formula for a 2-D hyperplane is given as follows:

$$max(W(\alpha)) = \Sigma_{i=1}^{m}\alpha_i - \frac{1}{2}(\Sigma_{i=1}^{m}\Sigma_{j=1}^{m}\alpha_i\alpha_j y_i y_j x_i^T x_j) \quad (4)$$

where,
vectors that lie on the boundary having $\alpha > 0$ are called support vectors

e) SGD Classifier: A linear classifier optimized by the stochastic gradient descent technique is known as the SGD Classifier. It has been used in this research to optimize the SVM and Logistic Regression models. The model predicts a cost for each sample that is inaccurately predicted and hence improves the accuracy of the prediction by trying various coefficient values and searching for the one that minimizes the cost margin. This calculation is used to update the coefficient in the next iteration using a pre-defined learning rate. A notable characteristic of SGD is that the coefficient is updated for each sample in the training dataset and not at the end of the entire iteration over all rows of the training data. The learning rate must be decided based on which value of α gives the lowest training errors on the subsamples and then gradually decreased over time ([21]).

The following formula shows the SGD optimization on SVM -

$$\begin{aligned} \nabla_\beta h(\beta) = \beta, if 1 - y_i(_i) \leq 0 \\ else, \beta - C * y_i x_i \end{aligned} \quad (5)$$

where,
β = random co-efficient values
(x_i, y_i) are points on which change in gradient is checked

f) Extreme Gradient Boost: The XGBoost algorithm is an optimization on decision trees where the tree is generated sequentially. Each independent variable is given a weight and is then passed as input to the decision tree which forecasts an outcome. The variables are then fed into the second decision tree with a changed and enhanced weight for variables that were incorrectly classified previously. This boosting technique is known for its high accuracy, quick learning and provides effective memory management even on sparse datasets. The new models created predict errors of the previous models and then add them together make a final prediction([22]). However, it required complex calculations and more memory for training.

The boosting in XGBoost is given as follows -

$$w_j^* = \frac{G_j}{H_j + \lambda} \quad (6)$$

$$obj* = -\frac{1}{2}\Sigma_{j=1}^{T}\frac{G_j^2}{H_j + \lambda} + \gamma T \quad (7)$$

where,
γ is pruning parameter

3) Unsupervised Learning Models: Unsupervised learning models are often used to categorize unlabelled data and extract useful features. Supervised learning becomes expensive and time-consuming when there are multiple different feature spaces in a heterogeneous text corpus. Hence, unsupervised learning is suggested for big data applications [23].

a) K-Means Clustering: This unsupervised clustering algorithm groups data into clusters based on the similarities or differences in their characteristics. Most often, cluster centres are assigned in a random way and the remaining data/objects are allocated to the closest clusters. A distance measure is used to iteratively re-assign the different data points to the nearest clusters. The K-means algorithm tries to keep the inter-cluster distance maximum and intra-cluster distance minimum. Since K-means is sensitive to the initialization of the centroid points, the clusters were initialized using the "k-means++" parameter in the model such that they remain far apart from each other. The objective of K-means is to minimize the following objective function, i.e.., within-cluster sum -

$$argmin\Sigma_{j=1}^{k}\Sigma_{x\epsilon S_i}||x - \mu_i||^2 \quad (8)$$

where,
$(x_1, x_2..., x_n)$ are the set of observations
μ is the cluster center

The clustering algorithm was implemented using TF-IDF and Word2Vec word embedding models which were compared to find better performance.

b) Latent Dirichlet Allocation: The LDA model is a generative, probabilistic model that uses word probabilities to display topics in a corpus. The topic is defined by a distribution of words which is represented as a random mixture of latent themes. Every document is represented as a probabilistic distribution across latent topics. According to the LDA premise, each document is represented as a probabilistic distribution across latent topics. Every latent topic represented is also represented as a probabilistic word distribution, and the word distributions of the topics all have the same Dirichlet prior. LDA is also used for dimensionality reduction and performs well while mixing distributional documents from various topics both inside and outside of each other. The total probability for the LDA model is given by -

$$\begin{aligned} P(W, Z, \theta, \varphi; \alpha, \beta) = \Pi_{i=1}^{K}P(\varphi_i; \beta)\Pi_{i=1}^{K}P(\theta_j; \alpha) \\ \Pi_{i=1}^{K}P(Z_{j,t}|\theta_j)P(W_{j,t}|\varphi z_{jt}) \end{aligned} \quad (9)$$

where,
$P(\varphi_i; \beta)$ = Dirichlet distribution of topics over terms
$P(\theta_j; \alpha)$ = Dirichlet distribution of documents over topics
$P(Z_{z,t}|\theta_j)$ = Probability of a topic appearing given a document
$P(W_{j,t}|\varphi z_{jt})$ = Probability of word appearing given a topic

The LDA method was used with TF-IDF word embeddings. The pre-processed documents were passed through the doc2bow model to convert the words to the bag-of-words format which is then passed to TfIDfVectorizer to generate TF-IDF word embeddings.

4) Pre-Trained Models: Pre-Trained models have various advantages and applications as they require less training time compared to custom-built models, easier to implement, and give high accuracy. They are trained on large datasets to perform specific tasks. The used models are specifically designed to perform sentiment analysis on textual data.

a) VADER: VADER stands for Valence Aware Dictionary and Sentiment Reasoner. It is an open-source, rule-based tool for sentiment analysis. It is predominantly trained on social media data and is able to recognize acronyms, western-style emoticons and commonly used slang. The VADER model indicates whether a vocabulary is positive, negative, or neutral. We receive a Python dictionary containing four keys and their matching values as the output from VADER. Negative, Neutral, Positivity, and Compound are the acronyms for these three terms. The other 3 scores (neg, neu, and pos) are normalized between -1 and +1 to get the Compound score.

b) TextBlob: TextBlob present in the NLTK Python library is rule-based and necessitates a pre-defined list of categorised terms. It uses a word and weight dictionary, which has some scores that assist in determining the polarity of a statement. Sentiments are determined considering the frequency of each word in an input sentence and the semantic relationships between those words. The polarity is represented on scale [-1,1]. TextBlob uses "intensity" to calculate semantic and subjective measures.

C. Model Evaluation

1) Accuracy: Percentage of predicted data that is correctly classified by the model in comparison to the correct labels in the texting dataset.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (10)$$

Since the dataset used in this study is highly skewed, this performance metric may not be an optimal way to judge the performance of the model, since equal weight is assigned to false positives and false negatives. For example, even if the model predicts all instances to belong to the majority class, the accuracy would still not be low even though we can clearly say that the model has poor performance.

2) Precision: Calculating how many of the instances that are supposed to be detected by the model as positives were actually detected.

$$Precision = \frac{TP}{TP + FP} \quad (11)$$

3) Recall: Detecting how many of the instances classified by the model as Positives, are actually True Positives.

$$Recall = \frac{TP}{TP + FN} \quad (12)$$

4) F1-Score: A combined metric (or) harmonic mean of precision and recall. Since the dataset is highly skewed, the weighted F1-score is the metric taken into consideration rather than the macro F1-score.

$$F1 = \frac{2 * Precision * Recall}{Precision + Recall} \quad (13)$$

5) Inertia: Inertia is a metric used to evaluate the K-means clustering method. It is the sum of squares of the distance between a data point and the centroid of its corresponding cluster.

$$Inertia = \Sigma_{i=1}^{N}(x_i - C_k)^2 \quad (14)$$

where,
N = number of samples within the dataset
C = cluster center

6) Distortion: Distortion is also a measure to evaluate the K-means clustering algorithm. It is the average of the Euclidian squared distance of all the samples to their respective cluster.

$$Distortion = \frac{1}{N}\Sigma_{i=1}^{N}\sqrt{(x_i - C_k)^2} \quad (15)$$

where,
N = number of samples within the dataset
C = cluster center

7) Coherence Value: This metric is used to evaluate the performance of the LDA model. Topic Coherence is an intrinsic metric of evaluation that scores every topic by assessing the degree of semantic similarity between high-scoring terms. It measures the similarity between the top N words which represent a topic. TThe measure is built around a sliding window, one-set segmentation of the top words, and an indirect confirmation measure based on normalised pointwise mutual information (NPMI) and cosine similarity.

VI. Results and Discussions

A. Supervised Learning Models

Impact of Hyper-parameter Tuning with GridSearchCV: TableI compares the supervised models before and after hyper-parameter tuning. Weighted F1-score is used as the evaluation metric. It is observed that in most models, there is an improvement in performance when the optimal parameters were used.

The comparison of performance based on various metrics is shown in TableII. For the dataset in consideration, decision trees have shown low performance due to highly dimensional and skewed data. This is because picking a set of attributes and calculating similarity scores are considered to be a challenge in decision trees. It can also be difficult to trace the steps and understand why a particular classification occurs. However, XGBoost which is an optimization on decision trees showed improvement. The XGBoost model can be parallelized to improve time efficiency despite the complexity of the algorithm. Although hyper-parameter tuning was performed, the default parameters gave a better F1 score.

The Multinomial Naïve Bayes(MNB) model showed better performance when compared to Decision Trees but was unable

TABLE I: Impact of Hyper-parameter Tuning on the Weighted F1-score of Supervised Learning Models

Model	Weighted F1-score (without Hyper-parameter Tuning)	Weighted F1-score (with Hyper-parameter Tuning)
Decision Tree	0.7077	0.7155
XG Boost	0.8242	0.8320
Multinomial Naïve Bayes (MNB)	0.7146	0.8163
Multinomial Logistic Regression (MLR)	0.8420	0.8472
SGD Optimization over MLR	0.8462	0.8477
Support Vector Machine (SVM)	0.8570	0.8516
SGD Optimization over SVM	0.8477	0.8514

TABLE II: Performance Metrics of The Implemented Supervised Models

Model	Accuracy Score	Recall	Precision	F1-score	Time Taken (in seconds)
Decision Tree	0.8002	0.6915	0.8002	0.7155	**6.091**
XG Boost	0.8461	0.8250	0.8438	0.8320	167.955
Multinomial Naïve Bayes (MNB)	0.8442	0.8083	0.8422	0.8163	10.870
Multinomial Logistic Regression(MLR)	0.8594	0.8410	0.8595	0.8472	113.678
SGD Optimization over MLR	0.8563	0.8422	0.8563	0.8477	11.724
Support Vector Machine(SVM)	**0.8614**	**0.8458**	**0.8615**	**0.8516**	3260.785
SGD Optimization over SVM	0.8559	0.8481	0.8559	0.8514	7.434

to capture the features of the highly dimensional dataset. The MNB model assumes independent features in the dataset and is quite simplistic. Due to this it avoids noise in the dataset and is unable to work well with complex features. The Multinomial Logistic Regression(MLR) performs fairly in comparison to the other models. It was effectively able to extract the important features of the data by using the model coefficients. However, overfitting of the data may be possible because the dataset being used is highly dimensional in nature. Hence, while hyper-parameter tuning, it was L2 regularized to avoid this scenario. This model was further optimized using the SGD Classifier. This is done by setting the loss parameter to 'log'. It showed a slight improvement in accuracy and a drastic improvement in speed. As the model is frequently updated based on the cost function, many oscillations are created to avoid convergence at the local minimum. Furthermore, only a single observation is processed at a time which makes it easier to fit into memory. The SVM model proved to be the best with a weighted F1-score of 0.8516. This is because SVM is an efficient model for noisy datasets [23]. Due to its generalization capabilities, it proved to show better performance as it did not overfit the data even when the input matrices were sparse. The kernel trick is a method that can be used to specify various linear and non-linear kernels in SVM to decide the shape of the hyperplane [6]. Non-linear kernels, especially 'rbf' and 'gaussian', upon experimentation, gave better results in comparison to 'linear' and 'polynomial' kernels.

The SGD Optimization over SVM performed almost as well as the SVM model itself but proved to be much better in terms of time complexity. The SGDClassifier with the 'hinge' loss parameter performs this optimization on SVM with a linear kernel, hence causing the slight drop in F1-score.

B. Unsupervised Learning Models

Due to the imbalanced nature of the data, K-means clustering did not seem to give the expected results because K-means forms spherical clusters and does not work well with clusters of other shapes. In this scenario, it would give higher weights to the bigger clusters. TableIII shows the comparison of the performance of the K-means algorithms using TF-IDF Weighting and Word2Vec(W2V) algorithm. The clustering with W2V embeddings showed better performance for this dataset. It gave smaller values for both inertia and distortion which indicates high inter-cluster distance and low intra-cluster distance. For the K-means model with W2V embeddings, the number of optimal clusters was estimated using the distortion metric. As seen in Fig.2, it was noticed that k=4 is the optimal number of clusters.

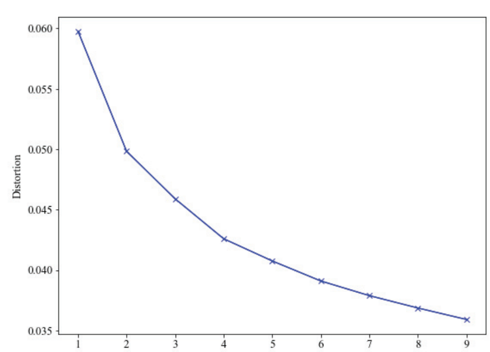

Fig. 2. Elbow Curve for K-Means Clustering with Word2Vec Embeddings to find the Optimal Number of Clusters

Amongst the unsupervised learning models, LDA showed better results than the K-means clustering model. The optimal

TABLE III: Performance of K-Means Clustering

Word Embeddings	Inertia	Distortion	Time Taken (in s)
TF-IDF	55648.10	0.9742	**11.25**
W2V	**174.27**	**0.0473**	187.91

number of topics for the LDA model was chosen to be $k = 4$ with the highest coherence value of 0.3889. After this point in the graph shown in Fig.3, there is a significant drop in coherence.

Although the coherence value for $k = 8$ is also high, there was a significant average topic overlap. The LDA model performs better than K-means but the overall performance is still quite low due to the skewness of the dataset. The LDA model also assumes unimodal Gaussian likelihood and hence does not preserve the complex features of the corpus needed for accurate classification.

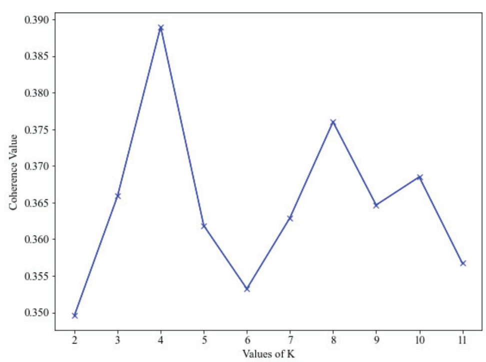

Fig. 3. Plot of Coherence Values for Different number of Topics for LDA Model

C. Pre-trained Models

When both the pre-trained models were compared, TextBlob shows slightly better performance than VADER. For both models, it was noticed that the raw text input resulted in better accuracy than the pre-processed text data. Since these models, especially VADER, are trained on social media data, often punctuation marks (like '!' or '...') or emojis are used to depict a sentiment. The models take these into consideration while calculating polarity values. Hence, removing such characters in the pre-processing has led to a change in polarity. Let us take an example of the following text input -

> The room is lovely with a scenic view from the balcony. EXCELLENT room service!!! Very happy with my stay in this hotel and would definitely visit again with my family :)

This text contained capital letters, exclamation marks and an emoji which all change the polarity of the sentiment detected by both VADER and TextBlob which was 0.96 and 0.60 respectively. However, for the following pre-processed input of the same review -

> room love scenic view balconi excel room servic happi stay hotel would definit visit famili

The polarities changed to 0.82 and 0.50 for VADER and TextBlob respectively and making the polarity less positive than the previous input. The performance of both these models is shown in TableIV.

TABLE IV: Performance of Pre-trained Models

Model	Weighted F1 Score (Raw Input)	Weighted F1 Score (Pre-processed Input)
VADER	0.7876	0.7570
TextBlob	**0.7991**	**0.7421**

VII. Conclusion

Sentiment Analysis is a domain of Natural Language Processing that is an efficient way to process unstructured text data and extract subjective information. It is applied in various domains especially to analyze user reviews and opinions on products and services. We aimed to employ this popular technique in the tourism domain for the purpose of analyzing user reviews on hotels. This helps the hotels improve their services by taking into consideration customer feedback and also helps future visitors decide if a particular hotel is a value for their money. The proposed study, while analyzing various supervised, unsupervised and pre-trained models, also gives an insight into what dataset features contributed to the success or failure of each model. The SVM supervised learning model gave the highest F1-score of 0.8516. In general, supervised learning models showed the best performance. The pre-trained models also faired quite well. The TextBlob model proved to be better than VADER model. However, the unsupervised learning models were unable to capture the skewed nature of the data and performed poorly in comparison to the rest.

VIII. Future Prospects

In this study, the most popular supervised, unsupervised and pre-trained models were used and the performance of each was analyzed specifically with respect to the characteristics of this hotel review dataset. In the pre-processing step, filtering untrustworthy data is an interesting option that can be considered to eliminate bias from the model. Most text reviews are short and it is challenging to extract features from such reviews to obtain accurate results. Hence, several dimensionality reduction techniques like PCA and SVD can be used to extract these important features of the data. Since hyper-parameter tuning with GridSearchCV is very time-consuming and requires high computational resources, other methods like RandomizedSearchCV can be used. This method is much faster as only a fixed number of parameters in the distribution are tried out. In rare cases, however, it may not lead to the optimal parameters as the search is not exhaustive. Clustering algorithms, like DBSCAN, that take into account the skewness of the data can be experimented with. Additionally, Transformer models like BERT and DomBERT can be explored.

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